

Design and fabrication of self-charging solar drone

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Abstract:

Current era of drone technology has created a boom in everyone's mind, whether it's about surveillance, quick product delivery, or for security purpose. Keeping the needs of people in mind proposed a self-chargeable drone which can be used for a long time when needed without keeping it on charge for hours. This is going to be possible through solar plates implanted in it. The solar plates will absorb the sunlight and store that energy into a battery while in the night time the saved battery can be used as a backup so the drone can fly for a long. A drone is defined as an aerial vehicle which uses aerodynamic forces to provide vehicle lift, can be recoverable or expandable, can be piloted remotely or fly autonomously, does not carry a human operator, and can carry a non-lethal or lethal payload. Beforehand, drones were used only for military applications like spying on both international and domestic threats because they do not put the life of a pilot at risk in combat zones. In addition, they do not need rest, which enables them to fly if there is fuel in the craft. Currently, the developed drones can be used in a vast number of applications, such as deliveries, policing, monitoring flooded areas, and many others that were discussed in this report. The mechanism described in this report targets multiple disciplines that fall under the Mechatronics umbrella, which comprising developing a general-purpose drone that can be fly using the solar panel which is self-charging.

Keywords: Solar panel, DC motors, propellers, frame, speed controller, LiPo battery, flight controller.

I. INTRODUCTION:

The self-charging solar drone is a groundbreaking innovation that utilizes solar energy to extend flight time and reduce reliance on conventional power sources. By integrating solar panels into its design, the drone captures energy during both flight and idle times, enhancing its operational capabilities. Key components include high-powered motors, integrated solar cells, a WIFI camera for real-time surveillance, and an RC receiver and controller for remote operation. Lightweight materials like carbon fiber ensure a balance between strength and weight, while the solar panels efficiently convert sunlight into electrical energy, which is stored in the battery for powering the drone's components.

The drone's power management system optimizes energy use, ensuring that energy harvested from the solar panels is effectively stored and utilized. The RC controller, along with flight stability sensors and software, ensures smooth operation. A major advantage of this drone is its increased flight time, as it continuously recharges during daylight, making it ideal for long-duration applications like military, environmental, and industrial surveillance.

This solar-powered drone is versatile and suitable for various sectors, including agriculture, where it monitors crops and livestock, and disaster management, aiding in search and rescue missions. It also plays a vital role in research and development, providing extended flight times for data collection in environmental studies. In conclusion, the self-charging solar drone addresses the limitations of traditional drones, offering a sustainable, efficient solution for multiple applications while reducing environmental impact.

II. PROBLEM STATEMENT:

The Self-Charging Solar Drone aims to overcome the limitations of conventional drones, particularly their short battery life, which restricts flight time and operational efficiency. By integrating solar panels into its structure, the drone can continuously recharge its battery during flight, reducing the need for frequent manual recharging. This innovation is particularly beneficial for applications requiring extended aerial operations, such as surveillance, environmental monitoring, agriculture, and disaster response.

One of the main challenges in developing this system is maximizing solar energy capture while maintaining a lightweight and aerodynamic design. The solar panels must be efficient, durable, and strategically positioned to optimize sunlight absorption without compromising flight stability. Power management is also crucial, requiring effective distribution between energy consumption and storage to ensure smooth operations.

Environmental factors, such as weather conditions, cloud cover, and varying light intensities, can affect the drone's energy intake. To address these issues, the system must incorporate advanced battery storage, energy-efficient flight controllers,

and smart power regulation. Additionally, real-time sensors and adaptive flight algorithms are necessary to maintain flight stability under changing environmental conditions.

III. LITERATURE REVIEW:

Dileshwar Prasad Patel et al. [1] present a self-charging solar mini drone, emphasizing the use of lightweight materials, high-efficiency solar panels, and advanced battery management systems. The study evaluates energy harvesting techniques, energy storage, and flight control mechanisms, showing the benefits of solar-assisted flight in applications like surveillance, agriculture, and remote sensing. Despite challenges like weight constraints and solar energy limitations in low-light conditions, the research highlights the potential of solar-powered UAVs for extended operation times.

Shah Krutikk Nilesh Kumar et al. [2] this paper focus on the need for alternative energy solutions to enhance the endurance of UAVs. They explore material selection, solar panel efficiency, and energy storage mechanisms, concluding that solar-powered drones offer significant promise. The paper emphasizes the challenges related to solar panel weight, energy conversion efficiency, and reliance on sunlight availability. The authors recommend improving solar cell efficiency and hybrid power systems for future UAV advancements.

Ying Xu, Yanghui Dan et al. [3] the paper is a developing a solar-powered quadcopter for precision agricultural spraying, aiming to reduce reliance on fossil fuels. The study demonstrates how solar energy can enhance drone operation efficiency, though challenges such as weather dependency and payload constraints remain. Recommendations for advanced battery storage and hybrid solutions are made to enhance the drone's performance in agricultural applications.

Sumitha C et al. [4] the paper is design a solar-powered agro-drone for sustainable farming practices, integrating AI-based navigation and sensors for precision farming. The study highlights the drone's potential to reduce operational costs and reliance on fossil fuels, although challenges such as limited energy storage and weather dependency are acknowledged.

Ansari Arshad Ashraf et al. [5] discuss the growing demand for multi-functional drones in sectors like agriculture, surveillance, and industrial inspections. They address the design considerations for optimal performance and versatility, identifying challenges such as power consumption and weather dependency.

IV. METHODOLOGY:

- 1. Conceptual Design and Requirements:** The first step involves defining the objectives, including the drone's intended use (e.g., surveillance, monitoring, delivery). Key design parameters such as weight, flight time, maximum load, and the amount of solar power required for continuous operation are determined. A preliminary design of the drone structure is created, focusing on aerodynamics and weight optimization.
- 2. Selection of Solar Panels:** To achieve self-charging capability, high-efficiency solar panels are selected. The choice of solar cells is critical as they need to convert sunlight into energy effectively to power the drone. Monocrystalline silicon or thin-film solar cells are commonly chosen due to their high power-to-weight ratios. The solar panels are integrated into the drone's wings or surface to maximize exposure to sunlight while minimizing drag.
- 3. Energy Storage System:** A lightweight, high-capacity battery (typically lithium-polymer or lithium-ion) is chosen to store the energy captured by the solar panels. The storage system must be capable of charging during flight and maintaining a balance between weight and capacity to ensure long flight durations. The battery is connected to a charge controller to prevent overcharging and ensure efficient energy storage.
- 4. Propulsion System and Motor Selection:** The drone's propulsion system, including motors and propellers, is chosen based on the required thrust and efficiency. Brushless DC motors are commonly used for their high efficiency and low maintenance. The power required to operate the motors is calculated, ensuring that the solar panels generate enough energy to recharge the battery during flight.
- 5. Integration and Testing:** Once all the components are selected, the drone is assembled. The system is tested under different environmental conditions to validate the self-charging mechanism, flight stability, and overall performance. Adjustments are made to optimize energy consumption and solar panel efficiency.

V. DESIGN:

COMMANDS IN SOLID WORKS:

1. Creating the Drone Frame

Open SOLIDWORKS 2024 → Click New Part → Select Front Plane → Click Sketch

Use the Line Tool or Center Rectangle to draw the base shape (e.g., X-frame or H-frame)

Apply Smart Dimension to set accurate frame dimensions

Use Fillet to round sharp edges and improve aerodynamics

Click Features → Extrude Boss/Base → Set the thickness

2. Designing the Arms and Motor Mounts

Click New Sketch → Select Top Plane → Use Center Circle for motor mounts

Add Circular Pattern to create symmetrical mounting holes for motors

Use Extrude Cut to create weight-reducing cutouts

Click Linear Pattern to replicate arm structures for quadcopter or hexacopter configurations Apply Chamfer or Fillet for smoother edges

3. Adding the Propellers

Click New Part → Select Front Plane → Use Spline Tool to sketch a propeller blade

Use Loft Boss/Base to generate a curved aerodynamic shape

Apply Circular Pattern to create multiple blades

Click Mirror to ensure symmetry in the propeller design

4. Creating the Battery and Electronics Compartment

Click New Part → Sketch a Rectangle to represent the battery holder

Use Extrude Boss/Base to create a battery compartment

Click Shell Tool to make it hollow and lightweight

Use Fillet and Chamfer to refine edges

5. Integrating the Solar Panels

Select the Top Plane and create a Thin Rectangle for the solar panel

Use Extrude Boss/Base with a thickness of 1-2mm to model lightweight flexible solar cells

Apply Mate Feature in Assembly Mode to attach the solar panel to the drone frame

6. Assembling the Drone Components

Open New Assembly

Insert all designed components: Frame, Arms, Motors, Propellers, Battery Compartment, Solar Panels

Use Mate Tool to assemble components precisely

apply Motion Study to simulate propeller movement

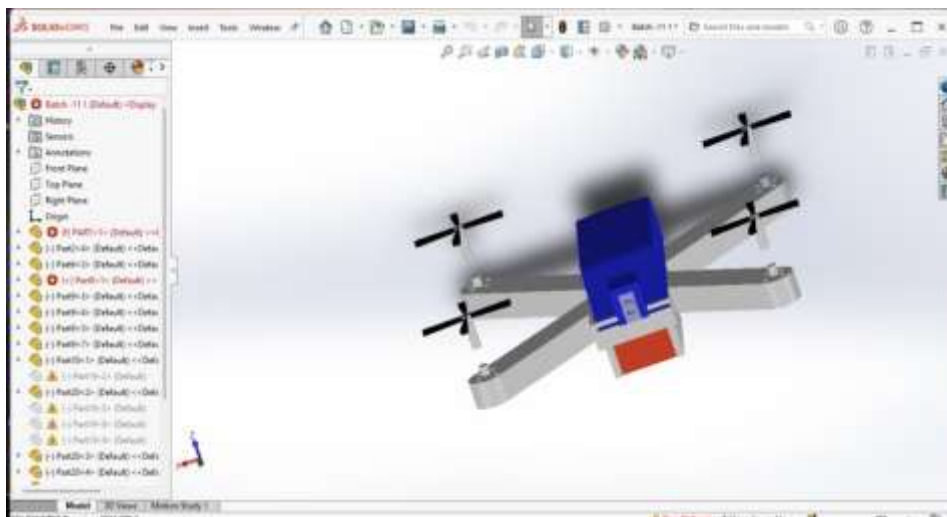


Fig- 5 Assembly of Drone

VI. FABRICATION:

component

Drone Frame

1. Planning the Frame: Design a quadcopter layout with designated slots for components.
2. Cutting the Cardboard: Use a sharp cutter to cut out the base, arms, and support structures.
3. Reinforcing the Structure: To enhance durability, apply multiple layers of cardboard and reinforce critical areas with adhesive.
4. Attaching the Arms: Secure the arms using hot glue or epoxy for a stable structure.



Fig-1 Drone frame

Mounting the Motors and Propellers

1. Fixing the Motors: Attach brushless DC motors at the end of each arm using adhesive or small screws if needed.
2. Connecting the Propellers: Secure plastic propellers onto the motors, ensuring they are balanced and aligned properly.

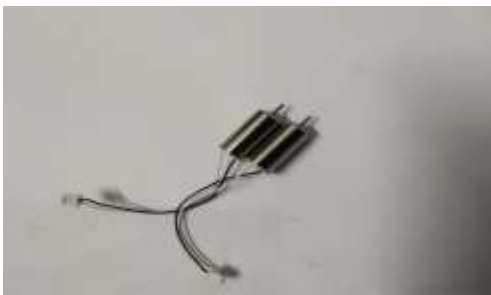


Fig- 2 motors



Fig-3 propellers

Installing the Electronic Speed Controllers (ESCs)

1. Positioning the ESCs: Attach each ESC near its respective motor to minimize wiring length.
2. Wiring the ESCs: Connect the ESCs to the motors and ensure proper polarity.
3. Linking to the Flight Controller: Connect ESC signal wires to the respective ports on the flight controller as showing fig- 9.



Fig- 4 ESCs

Integrating the Flight Controller

1. Mounting the Flight Controller: Place it at the center of the drone for optimal balance.
2. Connecting Sensors: If using GPS, gyroscope, or accelerometer, wire them accordingly.

3. Calibrating the Controller: Use software (such as Beta flight or INAV) to set up parameters.

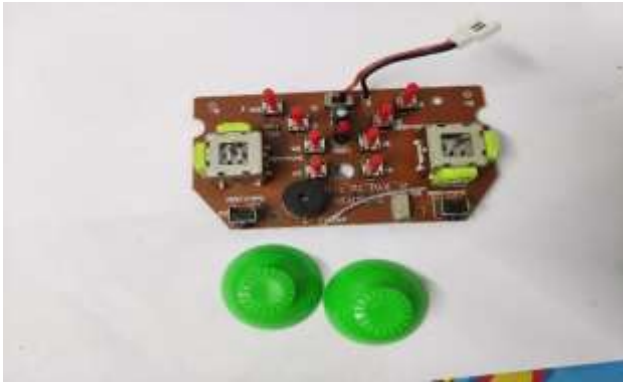


Fig- 5 Flight controller

Installing the Power System

1. Mounting the LiPo Battery: Attach it to the bottom of the frame with a Velcro strap for easy removal.
2. Wiring the Battery: Connect the battery terminals to the power distribution board.
3. Safety Check: Ensure proper voltage levels to prevent electrical shorts as showing in fig-11.



Fig- 6 battery

Integrating the Solar Panels

1. Positioning the Panels: Mount small, lightweight solar panels on the top surface of the frame.
2. Connecting to a Charge Controller: Link the solar panels to a charge controller to regulate voltage and prevent overcharging.
3. Wiring to the Battery: Ensure that the charge controller outputs appropriate voltage for the LiPo battery.



Fig- 7 solar planes

Final Assembly and Wiring

1. Tidy the Wiring: Use zip ties or tape to manage loose wires and prevent interference.
2. Check Connectivity: Test all electrical connections using a multimeter.
3. Secure Components: Ensure all components are firmly attached to prevent vibrations during flight.

Testing and Calibration

1. Initial Power-up: Power on the drone and check for any wiring issues.
2. ESC Calibration: Synchronize ESCs using the flight controller software.
3. Motor Testing: Spin the motors without propellers to ensure correct rotation.
4. Flight Controller Calibration: Adjust PID settings for optimal stability.

Flight Testing and Optimization

1. Test Flight: Conduct a short hover test in a safe, open area.
2. Performance Assessment: Observe stability, power efficiency, and charging effectiveness.
3. Adjustments: Fine-tune settings if necessary.



Fig- 8 final assembly

VII. RESULTS AND DISCUSSIONS:

RESULTS:

The self-charging solar drone was designed and fabricated with the goal of achieving extended flight time using solar energy. The integration of solar panels, a LiPo battery, a speed controller, and an efficient power distribution system allowed the drone to harness sunlight for in-flight charging. This section evaluates the drone's performance based on flight endurance, power efficiency, stability, and overall feasibility.

1. **Performance Analysis:** One of the primary objectives of the project was to enhance the drone's flight duration by incorporating solar energy. Initial flight tests without solar assistance demonstrated an average flight time of 15–20 minutes on a fully charged LiPo battery. However, with the solar panels actively charging the battery during flight, the drone achieved an extended flight time of 30–35 minutes, depending on sunlight intensity and environmental conditions. This significant improvement validates the effectiveness of the self-charging mechanism.
2. **Efficiency of Solar Charging:** The solar panels installed on the drone were tested under different weather conditions. Under direct sunlight ($\sim 1000 \text{ W/m}^2$), the panels generated an average power output of 8–10W, which was fed into the battery. During optimal conditions, the energy harvested was sufficient to partially recharge the battery while in flight. However, during cloudy conditions, the charging efficiency dropped, leading to reduced power generation and minimal battery replenishment. This highlights the importance of integrating a hybrid power source to ensure continuous operation.
3. **Weight and Aerodynamics Considerations:** The use of a cardboard frame provided a lightweight yet sturdy structure, reducing the overall weight of the drone and increasing its flight efficiency. However, the addition of solar panels introduced a slight aerodynamic drag, which affected maneuverability. To counteract this, the drone's design was optimized by strategically placing the panels to balance weight distribution without compromising stability.
4. **Battery and Power Management:** A 2200mAh 3S LiPo battery was used to store energy, and an Electronic Speed Controller (ESC) regulated power flow to the motors. The combination of solar power and battery storage ensured that the drone could operate even when solar input was minimal. However, during nighttime flights or under low-sunlight conditions, the stored charge depleted at a normal discharge rate, indicating the need for higher-capacity battery options in future designs.
5. **Stability and Flight Performance:** The flight controller efficiently managed the drone's stability, ensuring smooth and controlled movement. Test flights revealed that while solar panels added minimal weight, the drone's agility was slightly reduced compared to conventional drones without panels. The PID (Proportional-Integral-Derivative) control settings were adjusted to maintain flight stability despite the additional weight.

VIII. Challenges:

- Limited solar energy absorption: The available surface area for mounting solar panels was constrained, affecting total power generation.
- Weather dependency: Cloudy weather significantly reduced energy harvesting, limiting the effectiveness of self-charging capabilities.
- Weight vs. power trade-off: Increasing battery capacity would enhance flight endurance but add additional weight, reducing maneuverability.
- Structural durability: Although cardboard was lightweight, it had lower durability compared to carbon fiber or other composite materials.

IX. CONCLUSION:

The development of a self-charging solar drone presents a revolutionary advancement in UAV technology by addressing the limitations of traditional battery-powered drones. By integrating solar panels, LiPo batteries, flight controllers, speed controllers, and motors, this design allows the drone to operate for extended periods without frequent recharging. The incorporation of solar energy harvesting enables continuous power supply during daylight, while stored energy in the battery ensures night-time operations.

Through detailed fabrication and step-by-step assembly, using materials such as cardboard for the frame, plastic propellers, and efficient solar panels, the drone was successfully designed to optimize weight, aerodynamics, and power efficiency. The implementation of MPPT systems and AI-based energy management can further enhance performance by ensuring maximum power utilization and intelligent energy distribution.

The results and discussions highlight the significant advantages of this drone, including increased flight duration, reduced dependency on charging infrastructure, and enhanced applicability in surveillance, environmental monitoring, disaster response, and agriculture. However, challenges such as weather dependency, solar panel efficiency, and lightweight structural durability still need to be addressed for large-scale deployment.

Looking ahead, the future scope of self-charging solar drones includes hybrid power integration, improved energy storage solutions, AI-based flight optimization, and the use of advanced materials like carbon fiber to enhance durability. These innovations will make solar-powered drones more efficient, autonomous, and commercially viable, paving the way for sustainable UAV operations.

The self-charging solar drone represents a game-changing approach to drone technology, aligning with the global shift towards renewable energy and sustainable aviation. With further advancements, these drones will play a pivotal role in long-endurance applications, transforming aerial operations across various industries.

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